Scikit-learn or most commonly known as sklearn, is a machine learning library created by David Cournapeau in 2007 as part of his Google Summer of Code project. As mentioned, sklearn is a machine learning library specifically for the Python programming language. This library contains different functions and algorithms that are useful for creating data analysis and predictive models.

Sklearn contains many different features:

- First is data preprocessing, the library has the capability to clean data and pick relevant information for analysis accuracy.
- Second is its model evaluation capability, the library uses metrics and has a function that can be used to evaluate the performance of a model and has a model selection capability to find the best hyperparameters
- Third is pipeline support, this combines preprocessing and modeling steps and automates the workflow.
- Fourth is its capability to integrate with other python libraries such as pandas, numpy, and matplotlib. Integrating with these libraries makes it easier to analyze and visualize data.

Following that sklearn also has integrated algorithms that can be helpful in machine learning

Supervised Learning

This type of learning involves training models using data that has been labeled and considered the correct output. In classifications, sklearn uses and provides multiple algorithms when predicting categorical outcomes examples are logistic regression, decision trees, random forests, support vector machines (SVMs) and gradient boosting. While for regression models, which are models that predict continuous numerical values the library includes support for linear regression, support vector regression and decision tree regression.

Unsupervised Learning

This type of learning involves training models using data that is unlabeled that would be useful in discovering patterns and structures. First is clustering tasks, sklearn uses multiple algorithms to group similar data points. Examples would include K-means clustering, DBSCAN, and hierarchical clustering. Second is dimensionality-reduction,

this are models that handle data considered as high-dimensional. Sklearn has support for different techniques such as Principal Component Analysis (PCA).

Steps on how to install sklearn

In order to install and use sklearn, ensure you have Python set up on your machine.

You can easily install sklearn using pip, Python's standard package manager. Running this command will automatically pull in necessary dependencies, such as NumPy and Scipy:

```
[1]
16s
pip install -U scikit-learn

Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Collecting scikit-learn
  Downloading scikit_learn-1.8.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (11 kB)
Requirement already satisfied: numpy>=1.24.1 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (2.0.2)
Requirement already satisfied: scipy>=1.10.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Downloading scikit_learn-1.8.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (8.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 8.9/8.9 MB 23.8 MB/s eta 0:00:00
Installing collected packages: scikit-learn
  Attempting uninstall: scikit-learn
    Found existing installation: scikit-learn 1.6.1
    Uninstalling scikit-learn-1.6.1:
      Successfully uninstalled scikit-learn-1.6.1
Successfully installed scikit-learn-1.8.0
```

Once the process done, you can start using the library in your code by importing it:

```
[7]
0s
import sklearn
```

You can verify your installation by running this simple command in your Python:

```
[8]
0s
print(sklearn.__version__)

1.8.0
```

If you see an error, it means the package wasn't installed correctly. On the other hand, if a version number appears, you are all set.

Example Python Code

```python
import pandas as pd
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv(r"C:/Users/rochelle/Downloads/LORENZO_SAMPLE_UNSUPERVISED_Movies_Dataset.csv'

# Drop rows with missing values
df = df.dropna()

# Select numeric features for clustering (exclude non-numeric columns)
X = df[['Year',
        'Duration_Minutes',
        'IMDb_Rating',
        'Votes_Thousands',
        'Budget_Million_USD',
        'BoxOffice_Million_USD']]

# Scale features for better clustering performance
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Apply KMeans clustering
kmeans = KMeans(n_clusters=5, random_state=42, n_init=10)
df["Cluster"] = kmeans.fit_predict(X_scaled)

# Show results with movie titles
results = df[["Title", "Director", "Cluster", "IMDb_Rating", "BoxOffice_Million_USD"]]

# Print the full table
print("\nClustering Results (all rows):")
print(results.to_string(index=False))

```

```python
# Save to CSV
results.to_csv("clustering_output.csv", index=False)
print("\nTable saved as 'clustering_output.csv'")

# ----------------------------
# Plot the clusters using PCA
# ----------------------------

# Reduce dimensions to 2 for visualization
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

# Scatter plot of clusters
plt.figure(figsize=(8,6))
plt.scatter(X_pca[:,0], X_pca[:,1], c=df["Cluster"])
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Movie Clusters Visualization")
plt.show()
```

References

Codecademy. (2025). *Scikit-Learn Tutorial: Python Machine Learning Model Building*. Codecademy. https://www.codecademy.com/article/scikit-learn-tutorial

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